

Cloud Computing (Lab 13)

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BSE-V-B

LAB 13 – FULL FORMAT (Checklist)

TASK 0 – Lab Setup (Codespace & GH CLI)

Steps

1. Create repository and Codespace

```
gh repo create CC_<YourName>_<YourRollNumber>/Lab13 --public
```

```
gh codespace create --repo <username>/Lab13
```

```
gh codespace list
```

```
gh codespace ssh -c <codespace_name>
```

Screenshots

task0_codespace_create_and_list.png

task0_codespace_ssh_connected.png

TASK 1 – Create IAM Group

Steps

```
mkdir -p ~/Lab13
```

```
cd ~/Lab13
```

```
touch main.tf
```

```
terraform init
```

```
terraform apply -auto-approve
```

```
terraform output
```

Screenshots

task1_project_directory.png

task1_file_created.png

task1_main_tf.png

```
provider "aws" {
  shared_config_files = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

resource "aws_iam_group" "developers" {
  name = "developers"
  path = "/groups/"
}

output "group_details" {
  value = {
    group_name = aws_iam_group.developers.name
    group_arn = aws_iam_group.developers.arn
    unique_id = aws_iam_group.developers.unique_id
  }
}
```

task1_terraform_init.png

```
@toobashafique065 ~ /Lab13 $ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.27.0...
- Installed hashicorp/aws v6.27.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
@toobashafique065 ~ /Lab13 $
```

task1_terraform_apply.png

```
Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_iam_group.developers will be created
+ resource "aws_iam_group" "developers" {
  + arn      = (known after apply)
  + id       = (known after apply)
  + name     = "developers"
  + path     = "/groups/"
  + unique_id = (known after apply)
}

Plan: 1 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ group_details = {
  + group_arn = (known after apply)
  + group_name = "developers"
  + unique_id = (known after apply)
}
aws_iam_group.developers: Creating...
aws_iam_group.developers: Creation complete after 1s [id=developers]

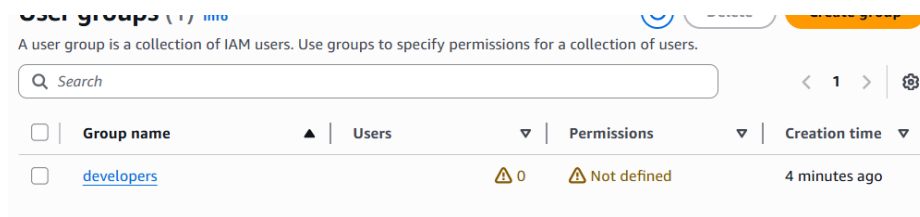
Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:
group_details = {
  "group_arn" = "arn:aws:iam::535942231766:group/groups/developers"
  "group_name" = "developers"
  "unique_id" = "AGPAXZSFF03LL76TSGUT"
}
```

task1_terraform_output.png

```
}
"arn" = "arn:aws:iam::535942231766:group/groups/developers"
"group_name" = "developers"
"unique_id" = "AGPAXZSFF03LL76TSGUT"
}
@toobashafique065 ~ /Lab13 $ terraform output
group_details = {
  "group_arn" = "arn:aws:iam::535942231766:group/groups/developers"
  "group_name" = "developers"
  "unique_id" = "AGPAXZSFF03LL76TSGUT"
}
```

task1_aws_console_group.png



TASK 2 – Create IAM User

Screenshots

task2_main_tf_user.png

```
}

resource "aws_iam_user" "lb" {
  name         = "loadbalancer"
  path         = "/users/"
  force_destroy = true
  tags = {
    DisplayName = "Load Balancer"
  }
}

resource "aws_iam_user_group_membership" "lb_membership" {
  user = aws_iam_user.lb.name
  groups = [
    aws_iam_group.developers.name
  ]
}

output "user_details" {
  value = {
    user_name = aws_iam_user.lb.name
    user_arn  = aws_iam_user.lb.arn
    unique_id = aws_iam_user.lb.unique_id
  }
}
```

task2_terraform_apply.png

```
Changes to Outputs:
+ user_details = {
+   unique_id = (known after apply)
+   user_arn  = (known after apply)
+   user_name = "loadbalancer"
}
aws_iam_user.lb: Creating...
aws_iam_user.lb: Creation complete after 1s [id=loadbalancer]
aws_iam_user_group_membership.lb_membership: Creating...
aws_iam_user_group_membership.lb_membership: Creation complete after 1s [id=terraform-20]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:
group_details = {
  "group_arn" = "arn:aws:iam::535942231766:group/groups/developers"
  "group_name" = "developers"
  "unique_id" = "AGPAXZSFF03LL76TSWGUT"
}
user_details = {
  "unique_id" = "AIDAXZSFF03LETVX6JFYN"
  "user_arn" = "arn:aws:iam::535942231766:user/users/loadbalancer"
  "user_name" = "loadbalancer"
}
```

task2_terraform_output.png

```

Outputs:

group_details = {
  "group_arn" = "arn:aws:iam::535942231766:group/groups/developers"
  "group_name" = "developers"
  "unique_id" = "AGPAXZSFF03LL76TSWGUT"
}
user_details = {
  "unique_id" = "AIDAXZSFF03LETVX6JFYN"
  "user_arn" = "arn:aws:iam::535942231766:user/users/loadbalancer"
  "user_name" = "loadbalancer"
}

```

task2_aws_console_user.png

User name	Path	Groups	Last activity	MFA
Lab1User	/	0	2 hours ago	-
LabUser	/	0	6 days ago	-
loadbalancer	/users/	1	-	-

task2_aws_console_user_groups.png

Group name	Users	Permissions
developers	1	Not defined

TASK 3 – Attach Policies

Screenshots

task3_main_tf_policies.png

```

# Attach AmazonEC2FullAccess policy to developers group
resource "aws_iam_group_policy_attachment" "developer_ec2_fullaccess" {
  group      = aws_iam_group.developers.name
  policy_arn = "arn:aws:iam::aws:policy/AmazonEC2FullAccess"
}

# Attach IAMUserChangePassword policy to developers group
resource "aws_iam_group_policy_attachment" "change_password" {
  group      = aws_iam_group.developers.name
  policy_arn = "arn:aws:iam::aws:policy/IAMUserChangePassword"
}

```

task3_terraform_apply.png

```

root@charlie005: ~/Lab13 $ terraform apply -auto-approve
aws_iam_group.developers: Refreshing state... [id=developers]
aws_iam_user.lb: Refreshing state... [id=loadbalancer]
aws_iam_user_group_membership.lb.membership: Refreshing state... [id=terraform-20260103190758479300000001]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
  + create

Terraform will perform the following actions:

# aws_iam_group_policy_attachment.change_password will be created
+ resource "aws_iam_group_policy_attachment" "change_password" {
  + group      = "developers"
  + id         = (known after apply)
  + policy_arn = "arn:aws:iam::aws:policy/IAMUserChangePassword"
}

# aws_iam_group_policy_attachment.developer_ec2_fullaccess will be created
+ resource "aws_iam_group_policy_attachment" "developer_ec2_fullaccess" {
  + group      = "developers"
  + id         = (known after apply)
  + policy_arn = "arn:aws:iam::aws:policy/AmazonEC2FullAccess"
}

Plan: 2 to add, 0 to change, 0 to destroy.
aws_iam_group_policy_attachment.change_password: Creating...
aws_iam_group_policy_attachment.developer_ec2_fullaccess: Creating...
aws_iam_group_policy_attachment.developer_ec2_fullaccess: Creation complete after 1s [id=developers-2026010319170421630
000001]

```

```

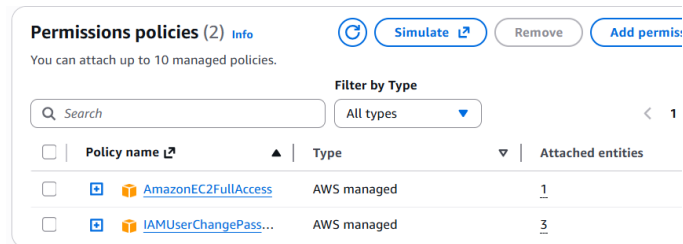
Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:

group_details = {
  "group_arn" = "arn:aws:iam::535942231766:group/groups/developers"
  "group_name" = "developers"
  "unique_id" = "AGPAXZSFF03LL76TSWGUT"
}
user_details = {
  "unique_id" = "AIDAXZSFF03LETVX6JFYN"
  "user_arn" = "arn:aws:iam::535942231766:user/users/loadbalancer"
  "user_name" = "loadbalancer"
}

```

task3_aws_console_policies.png



TASK 4 – Create Login Profile

Screenshots

task4_variables_tf.png

```

variable "iam_password" {
  description = "Temporary password for the IAM user"
  type        = string
  sensitive   = true
  default     = "IdontKnow"
}

```

task4_create_login_script.png

```

#!/usr/bin/env bash
set -euo pipefail

USERNAME="$1"
PASSWORD="$2"

# Check if login profile already exists
if aws iam get-login-profile --user-name "$USERNAME" >/dev/null 2>&1; then
  echo "Login profile already exists for $USERNAME. Skipping."
else
  echo "Creating login profile for $USERNAME"
  aws iam create-login-profile \
    --user-name "$USERNAME" \
    --password "$PASSWORD" \
    --password-reset-required
fi

```

task4_chmod_script.png

```

@toobashafique065 ~ /Lab13 $ chmod +x create-login-profile.sh
@toobashafique065 ~ /Lab13 $

```

task4_main_tf_login_profile.png

```

resource "aws_iam_user" "lb" {
  name         = "loadbalancer"
  path         = "/users/"
  force_destroy = true
  tags = {
    DisplayName = "Load Balancer"
  }
}

resource "null_resource" "create_login_profile" {
  triggers = {
    password_hash = sha256(var.iam_password)
    user          = aws_iam_user.lb.name
  }

  depends_on = [aws_iam_user.lb]

  provisioner "local-exec" {
    command = "${path.module}/create-login-profile.sh ${aws_iam_user.lb.name} '${var.iam_password}'"
  }
}

```

task4_terraform_apply.png

```

@toobashafique065 ~ ~/Lab13 $ terraform apply -auto-approve -var="iam_password=MySecurePass123!"
aws_iam_group.developers: Refreshing state... [id=developers]
aws_iam_user.lb: Refreshing state... [id=loadbalancer]
aws_iam_group_policy_attachment.change_password: Refreshing state... [id=developers-20260103191704218300000002]
aws_iam_group_policy_attachment.developer_ec2_fullaccess: Refreshing state... [id=developers-20260103191704216300000001]
aws_iam_user_group_membership.lb_membership: Refreshing state... [id=terraform-20260103190758479300000001]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
  + create

Terraform will perform the following actions:

# null_resource.create_login_profile will be created
+ resource "null_resource" "create_login_profile" {
  + id          = (known after apply)
  + triggers = {
    + "password_hash" = (sensitive value)
    + "user"          = "loadbalancer"
  }
}

Plan: 1 to add, 0 to change, 0 to destroy.
null_resource.create_login_profile: Creating...
null_resource.create_login_profile: Provisioning with 'local-exec'...
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile (local-exec): (output suppressed due to sensitive value in config)
null_resource.create_login_profile: Creation complete after 6s [id=7532446797447967914]

```

task4_aws_cli_verify.png

```

@toobashafique065 ~ ~/Lab13 $ aws iam get-login-profile --user-name loadbalancer
{
  "LoginProfile": {
    "UserName": "loadbalancer",
    "CreateDate": "2026-01-03T19:27:31+00:00",
    "PasswordResetRequired": true
  }
}

```

task4_aws_console_login.png

Account ID or alias ([Don't have?](#))

535942231766

☐ Remember this account

IAM username

loadbalancer

Password

.....

☐ Show Password [Having trouble?](#)

Sign in

Sign in using root user email

[Create a new AWS account](#)

task4_aws_console_password_reset.png

Your account (535942231766) password has expired or requires a reset.

To continue, please verify your old and set a new password for **loadbalancer** ([not you?](#)).

Old Password

.....

☐ Show Password

New Password

.....

Confirm New Password

.....

☐ Show Password Matches

Confirm Password Change

✓ Password reset successful

Password reset

Your IAM User (**loadbalancer**) password has been reset successfully.

Continue to sign in

TASK 5 – Create Access Keys

Screenshots

task5_main_tf_access_keys.png

```
}
resource "aws_iam_access_key" "lb_access_key" {
  user = aws_iam_user.lb.name
}

output "access_key_id" {
  value = aws_iam_access_key.lb_access_key.id
}

output "access_key_secret" {
  value = aws_iam_access_key.lb_access_key.secret
  sensitive = true
}
```

task5_terraform_apply.png

```
@toobashafique065 ~ ~/Lab13 $ terraform apply -auto-approve -var="iam_password=MySecurePass123!"
aws_iam_group.developers: Refreshing state... [id=developers]
aws_iam_user.lb: Refreshing state... [id=loadbalancer]
aws_iam_group_policy_attachment.change_password: Refreshing state... [id=developers-2026010319170421830000000]
aws_iam_group_policy_attachment.developer_ec2_fullaccess: Refreshing state... [id=developers-2026010319170421830000000]
null_resource.create_login_profile: Refreshing state... [id=7532446797447967914]
aws_iam_user_group_membership.lb_membership: Refreshing state... [id=terraform-20260103190758479300000001]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_iam_access_key.lb_access_key will be created
+ resource "aws_iam_access_key" "lb_access_key" {
  + create_date           = (known after apply)
  + encrypted_secret      = (known after apply)
  + encrypted_ses_smtp_password_v4 = (known after apply)
  + id                   = (known after apply)
  + key_fingerprint       = (known after apply)
  + secret               = (sensitive value)
  + ses_smtp_password_v4 = (sensitive value)
  + status               = "Active"
  + user                 = "loadbalancer"
}

Plan: 1 to add, 0 to change, 0 to destroy.
```

task5_terraform_output.png

```
Outputs:
access_key_id = "AKIAXZSFF03LF6GL4TZP"
access_key_secret = <sensitive>
group_details = {
  "group_arn" = "arn:aws:iam::535942231766:group/groups/developers"
  "group_name" = "developers"
  "unique_id" = "AGPAXZSFF03LL76TSWGUT"
}
user_details = {
  "unique_id" = "AIDAXZSFF03LETVX6JFYVN"
  "user_arn" = "arn:aws:iam::535942231766:user/users/loadbalancer"
  "user_name" = "loadbalancer"
}
```

task5_tfstate_secret.png

```
@toobashafique065 ~ ~/Lab13 $ cat terraform.tfstate | grep -A 10 "access_key_secret"
  "access_key_secret": {
    "value": "/Hp8a1S+aFi0myhhSPxvTzdIf63S0Ti27klumJqX",
    "type": "string",
    "sensitive": true
  },
  "group_details": {
    "value": {
      "group_arn": "arn:aws:iam::535942231766:group/groups/developers",
      "group_name": "developers",
      "unique_id": "AGPAXZSFF03LL76TSWGUT"
    }
  },
}
```

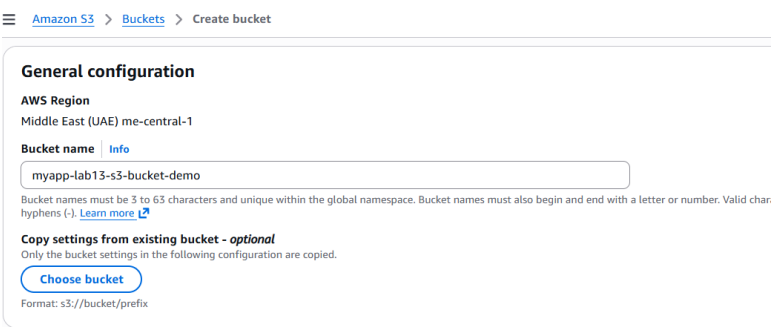

task5_aws_console_access_keys.png



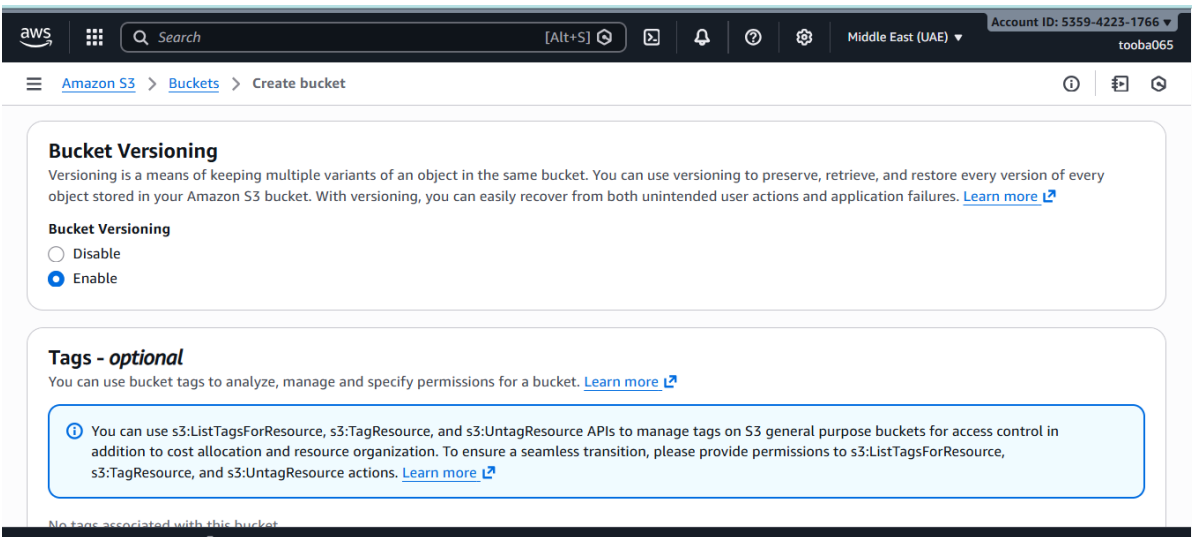
TASK 6 – Terraform Remote State (S3)

Screenshots

task6_s3_bucket_create.png



task6_s3_bucket_versioning.png



task6_main_tf_backend.png

```

terraform {
  backend "s3" {
    bucket = "myapp-lab13-s3-bucket-demo"
    key    = "myapp/terraform.tfstate"
    region = "me-central-1"
    encrypt = true
    use_lockfile = true
  }
}

```

task6_terraform_init_migrate.png

```

@toobashafique065 @ ~/Lab13 $ terraform init -migrate-state
Initializing the backend...
Do you want to copy existing state to the new backend?
  Pre-existing state was found while migrating the previous "local" backend to the
  newly configured "s3" backend. No existing state was found in the newly
  configured "s3" backend. Do you want to copy this state to the new "s3"
  backend? Enter "yes" to copy and "no" to start with an empty state.

  Enter a value: yes

Successfully configured the backend "s3"! Terraform will automatically
use this backend unless the backend configuration changes.
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Reusing previous version of hashicorp/null from the dependency lock file
- Using previously-installed hashicorp/aws v6.27.0
- Using previously-installed hashicorp/null v3.2.4

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
@toobashafique065 @ ~/Lab13 $

```

task6_terraform_apply.png

```

@toobashafique065 @ ~/Lab13 $ terraform apply -auto-approve -var="iam_password=MySecurePass123!"
aws_iam_group.developers: Refreshing state... [id=developers]
aws_iam_user.lb: Refreshing state... [id=loadbalancer]
null_resource.create_login_profile: Refreshing state... [id=7532446797447967914]
aws_iam_access_key.lb_access_key: Refreshing state... [id=AKIAXZSFF03LF6GL4TZP]
aws_iam_user_group.membership.lb_membership: Refreshing state... [id=terraform-202601031907584793000000001]
aws_iam_group_policy_attachment.change_password: Refreshing state... [id=developers-202601031917042183000000002]
aws_iam_group_policy_attachment.developer_ec2_fullaccess: Refreshing state... [id=developers-202601031917042163000000001]

No changes. Your infrastructure matches the configuration.

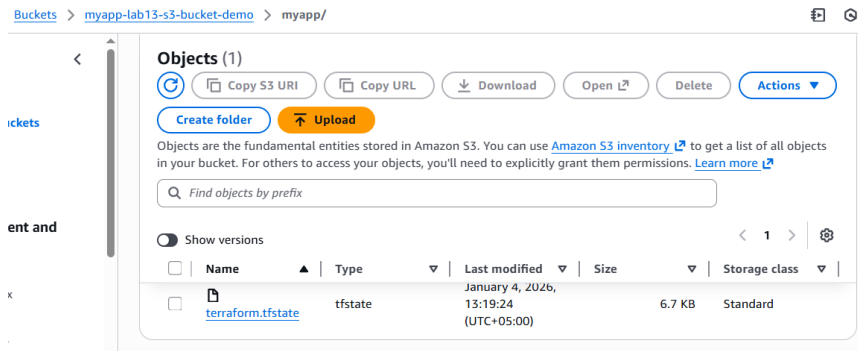
Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are
needed.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

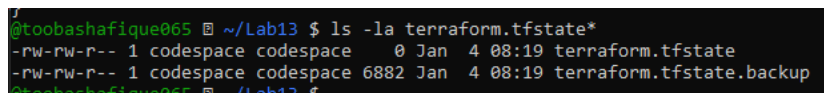
Outputs:
access_key_id = "AKIAXZSFF03LF6GL4TZP"
access_key_secret = <sensitive>
group_details = {
  "group_arn" = "arn:aws:iam::535942231766:group/groups/developers"
  "group_name" = "developers"
  "unique_id" = "AGPAXZSFF03LL76TSGUT"
}
user_details = {
  "unique_id" = "AIDAXZSFF03LETX6JFYV"
  "user_arn" = "arn:aws:iam::535942231766:user/users/loadbalancer"
  "user_name" = "loadbalancer"
}

```

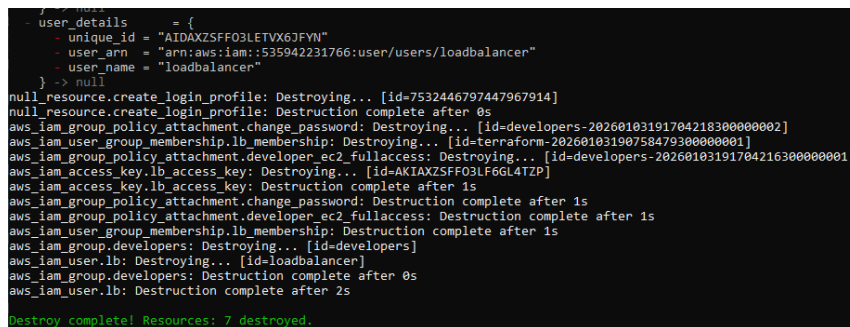
task6_s3_tfstate_file.png



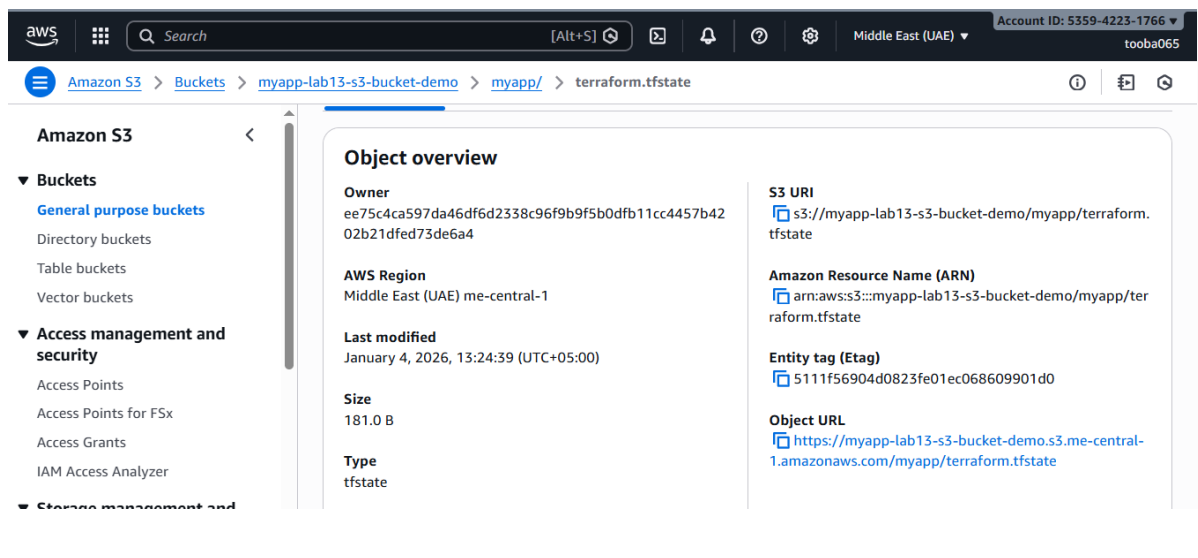
task6_local_state_backup.png



task6_terraform_destroy.png



task6_s3_tfstate_destroyed.png



TASK 7 – Multiple Users via CSV

Screenshots

task7_locals_tf.png

```
GNU nano 7.2
locals {
  users = csvdecode(file("users.csv"))
}
```

task7_users_csv.png

```
GNU nano 7.2
user_name
Michael
Dwight
Jim
Pam
Ryan
Andy
Robert
Stanley
Kevin
Angela
Oscar
Phyllis
Toby
Kelly
Darryl
Creed
Meredith
Erin
Gabe
Jan
David
Holly
Charles
Jo
Clark
```

task7_main_tf_multiple_users.png

```
GNU nano 7.2
terraform {
  backend "s3" {
    bucket     = "myapp-lab13-s3-bucket-demo"
    key        = "myapp/terraform.tfstate"
    region     = "us-east-1"
    encrypt    = true
    use_lockfile = true
  }
}

provider "aws" {
  shared_config_files = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

# Developers Group
resource "aws_iam_group" "developers" {
  name = "developers"
  path = "/groups/"
}

# Group Outputs
output "group_details" {
  value = {
    group_name = aws_iam_group.developers.name
    group_arn  = aws_iam_group.developers.arn
    unique_id  = aws_iam_group.developers.unique_id
  }
}

# Attach Policies to Group
resource "aws_iam_group_policy_attachment" "developer_ec2_fullaccess" {
```

task7_terraform_init.png

```

stoobashafique065 @ ~/Lab13 $ terraform init
Initializing the backend...
Initializing provider plugins...
- Reusing previous version of hashicorp/null from the dependency lock file
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/null v3.2.4
- Using previously-installed hashicorp/aws v6.27.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
stoobashafique065 @ ~/Lab13 $

```

task7_terraform_apply.png

```

stoobashafique065 @ ~/Lab13 $ terraform apply -auto-approve -var="iam_password=MySecurePass123!"
Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
+ create

Terraform will perform the following actions:

# aws_iam_access_key.users_access_keys["Andy"] will be created
+ resource "aws_iam_access_key" "users_access_keys" {
+   create_date           = (known after apply)
+   encrypted_secret      = (known after apply)
+   encrypted_ses_smtp_password_v4 = (known after apply)
+   id                   = (known after apply)
+   key_fingerprint       = (known after apply)
+   secret                = (sensitive value)
+   ses_smtp_password_v4  = (sensitive value)
+   status                = "Active"
+   user                  = "Andy"
}

# aws_iam_access_key.users_access_keys["Angela"] will be created
+ resource "aws_iam_access_key" "users_access_keys" {
+   create_date           = (known after apply)
+   encrypted_secret      = (known after apply)
+   encrypted_ses_smtp_password_v4 = (known after apply)
+   id                   = (known after apply)
+   key_fingerprint       = (known after apply)
+   secret                = (sensitive value)
+   ses_smtp_password_v4  = (sensitive value)
+   status                = "Active"
+   user                  = "Angela"
}

# aws_iam_access_key.users_access_keys["Charles"] will be created
+ resource "aws_iam_access_key" "users_access_keys" {

```

task7_terraform_output.png

```

@toobashafique065 ~ /Lab13 $ terraform output
all_access_key_secrets = <sensitive>
all_users_details = {
  "Andy" = {
    "access_key_id" = "AKIAZXSF03LNQ2P2MGS"
    "user_arn" = "arn:aws:iam::535942231766:user/users/Andy"
    "user_unique_id" = "AIDAXZSFF03LH6SVQG5X"
  }
  "Angela" = {
    "access_key_id" = "AKIAZXSF03LM75WKJOP"
    "user_arn" = "arn:aws:iam::535942231766:user/users/Angela"
    "user_unique_id" = "AIDAXZSFF03LI5D55QPDW"
  }
  "Charles" = {
    "access_key_id" = "AKIAZXSF03LCSBRYEUQ"
    "user_arn" = "arn:aws:iam::535942231766:user/users/Charles"
    "user_unique_id" = "AIDAXZSFF03LIMPASR6HF"
  }
  "Clark" = {
    "access_key_id" = "AKIAZXSF03LNFVFFIYC"
    "user_arn" = "arn:aws:iam::535942231766:user/users/Clark"
    "user_unique_id" = "AIDAXZSFF03LF2M2MRW2E"
  }
  "Creed" = {
    "access_key_id" = "AKIAZXSF03LAI4VF72N"
    "user_arn" = "arn:aws:iam::535942231766:user/users/Creed"
    "user_unique_id" = "AIDAXZSFF03LETY3PCLH4"
  }
  "Darryl" = {
    "access_key_id" = "AKIAZXSF03LAUZIAAVR"
    "user_arn" = "arn:aws:iam::535942231766:user/users/Darryl"
    "user_unique_id" = "AIDAXZSFF03LLIDXT2475"
  }
  "David" = {
    "access_key_id" = "AKIAZXSF03LIET6IJQS"
    "user_arn" = "arn:aws:iam::535942231766:user/users/David"
    "user_unique_id" = "AIDAXZSFF03LH7RQFOAJ6"
  }
  "Dwight" = {
    "access_key_id" = "AKIAZXSF03LLOQQ2WCJ"
    "user_arn" = "arn:aws:iam::535942231766:user/users/Dwight"
    "user_unique_id" = "AIDAXZSFF03LEXWE7JSEF"
  }
}

```

task7_tfstate_secrets.png

```

@toobashafique065 ~ /Lab13 $ cat terraform.tfstate | grep -A 5 "all_access_key_secrets"
@toobashafique065 ~ /Lab13 $

```

task7_aws_console_all_users.png

Users (28) Info 🔄 Delete

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Q Search < 1

☐	User name	Path	Group	Last activity
☐	Andy	/users/	1	-
☐	Angela	/users/	1	-
☐	Charles	/users/	1	-
☐	Clark	/users/	1	-
☐	Creed	/users/	1	-

task7_aws_console_group_members.png

User groups > developers

nd Access ent (IAM)

Users (26) Permissions Access Advisor

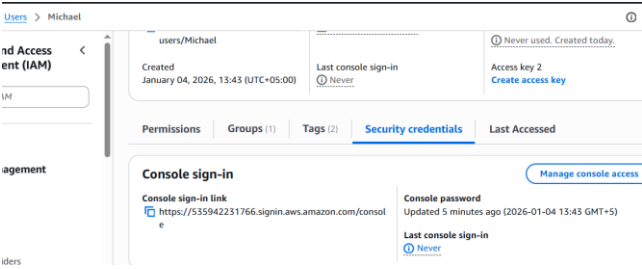
Users in this group (26) 🔄 Remove

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

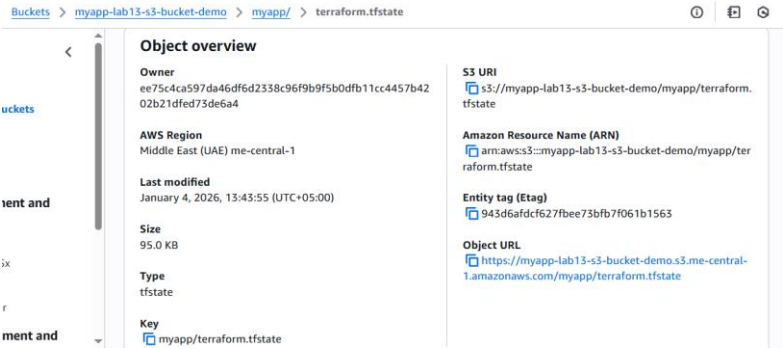
Q Search < 1

☐	User name
☐	Andy
☐	Angela
☐	Charles

task7_aws_console_user_access_key.png



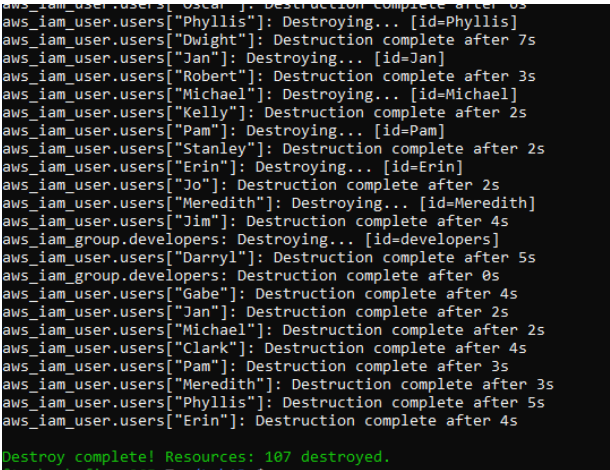
task7_s3_tfstate_multiple_users.png



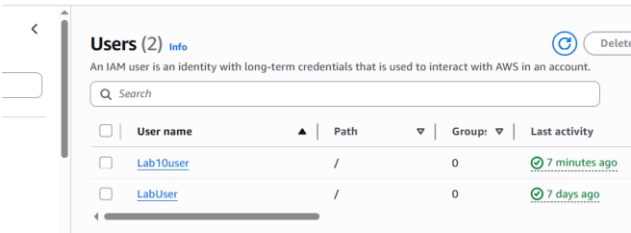
CLEANUP

Screenshots

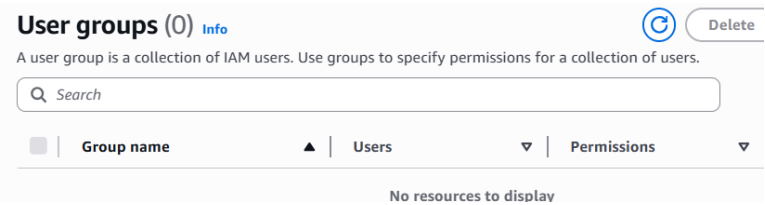
cleanup_destroy_complete.png



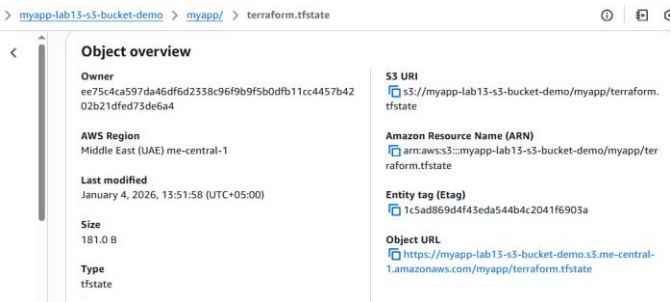
cleanup_aws_console_users_deleted.png



cleanup_aws_console_group_deleted.png



cleanup_s3_empty_state.png



cleanup_final_files.png

